Rotary Wing Aerodynamics (AE 667)



Applying BEM to Wind Turbines

Approach similar to that for helicopter rotors

- Divide blade into thin strips
- Compute lift, drag, etc. at the strips
- Integrate values at the strips to get the total forces & power.



New Terms / Terms Redefined

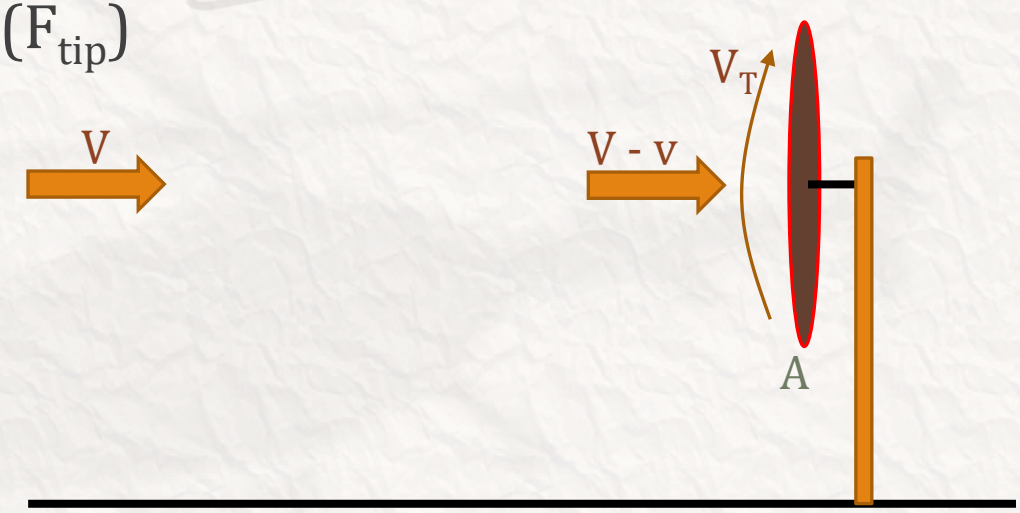
- **Induction Factor** (a)

  $a = v/V$

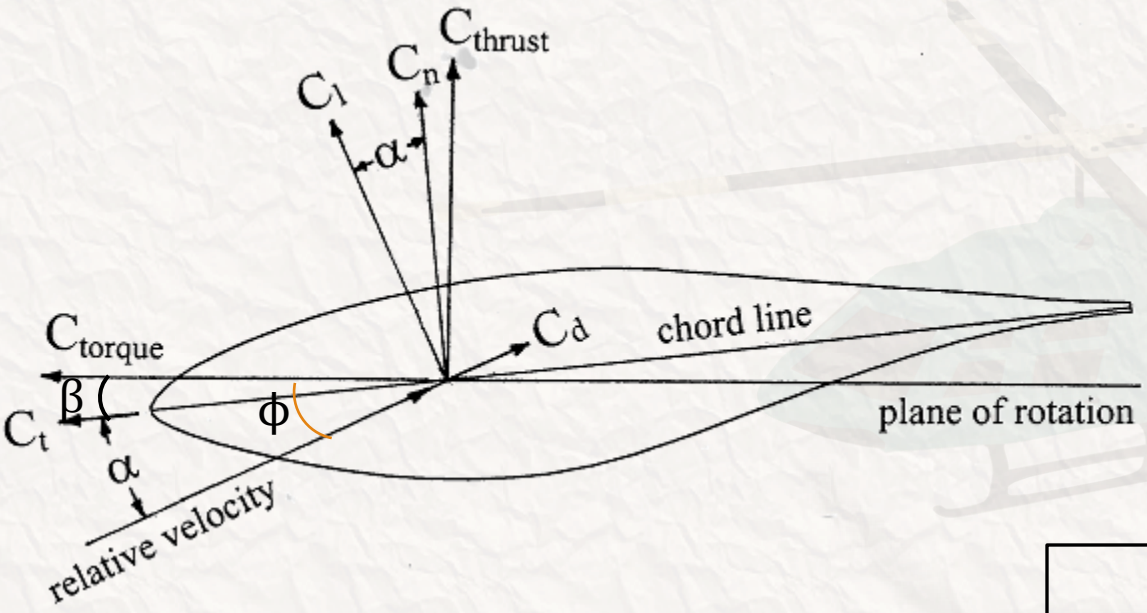
- **Tangential Velocity** (V_T)
- **Swirl Induction Factor** (a'):

  $V_T = r\Omega(1+a')$

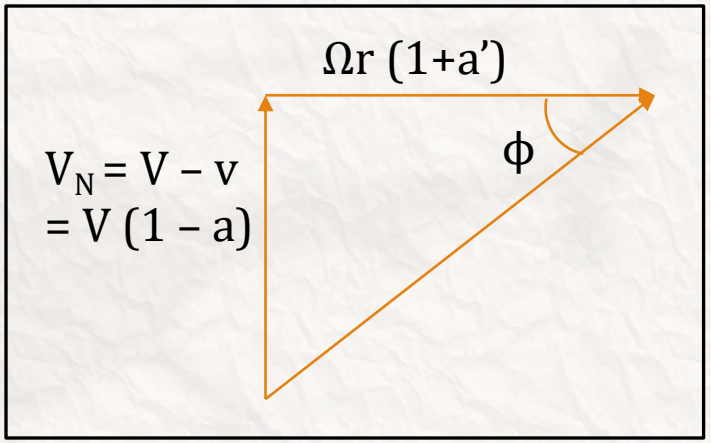
- **Tip loss correction factor** (F_{tip})
- $\lambda = \Omega r/V$



Velocities and Forces at Blades



V



Procedure

1. Obtain airfoil data (C_l, C_d, C_m) at angle of attack range of 0° to 359° at reasonable intervals
2. Divide the blade into several segments
3. Obtain angle of blade segments from rotor plane (β) at the center of each segment through information on pitch angle and blade twist.
4. Obtain chord lengths (c) at the center of each segment through information on taper.

Procedure

5. Initialize induction factors

$$a = \frac{1}{4}\left[2 + \pi\lambda\sigma - \sqrt{4 - 4\pi\lambda\sigma + \pi\lambda^2\sigma\,(8\beta + \pi\sigma)}\right]; a' = 0;$$

6. Compute $V_N = V(1 - a)$; $V_T = r\Omega(1 + a')$

7. Effective angle of wind w.r.t blade

$$\phi = \tan^{-1}\left(\frac{V_N}{V_T}\right)$$

8. Effective angle of attack at the blade segments:

$$\alpha = \phi - \beta$$

9. α is used to find C_l, C_d at each segment from airfoil tables

Procedure

10. Tip loss factor is computed as:

$$F_{tip} = \frac{2}{\pi} cos^{-1}[e^{\frac{B}{2}\left(\frac{R-r}{r \sin \phi}\right)}]$$

11. Updated values for induction factors are computed:

$$a = \left[1 + \frac{4F \sin^2 \phi}{\sigma(C_l \cos \phi + C_d \sin \phi)}\right]^{-1}$$

$$a' = \left[-1 + \frac{4F \sin \phi \cos \phi}{\sigma(C_l \cos \phi - C_d \sin \phi)}\right]^{-1}$$

12. The updated induction factors are used in step number 6 and the steps 6 – 11 are iterated until the induction factors converge.

Procedure

13. Lift and drag per unit span at each radial location is found using the obtained C_l and C_d, and the relative velocity

$$L' = \frac{1}{2}\rho\left[{V_N}^2 + {V_T}^2\right].c.C_l \qquad D' = \frac{1}{2}\rho\left[{V_N}^2 + {V_T}^2\right].c.C_d$$

14. In-plane force due to each blade segment is calculated by multiplying the effective force along rotor plane with the blade segment length

$$T' = dr[L' \sin\phi - D' \cos\phi]$$

15. Power generated by each blade segment is obtained by multiplying the segment speed to the in-plane segment force

$$P' = r\Omega T'$$

Procedure

16. Total Power generated is just summation of power generated by each blade segments times the number of blades

$$P_{Tot} = B \times \sum P'$$

17. Normal force coefficient is obtained by projection of lift and drag coefficients in direction normal to the local blade chord

$$C_n = C_l \cos\alpha + C_d \sin\alpha$$

18. Tangential force coefficient is obtained by projection of lift and drag coefficients in direction tangential to the local blade chord

$$C_t = C_l \sin\alpha - C_d \cos\alpha$$

Reflections…



https://forms.gle/7Fi8rZYJprUxHeVc6